



all

PROJECT IDS-WSN



INTRUSION DETECTION SYSTEM IN WIRELESS SENSOR NETWORKS



CONTENT OVERVIEW



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Dataset link – <https://www.kaggle.com/datasets/bassamkasasbeh1/wsnds>

UNDERSTANDING THE DATA

Checking the shape (value counts), info, column names, dtypes of columns, descriptive statistics, sample dataset etc of the data

► Dataset shape: (374661, 19)

Columns & dtypes:

```
id          int64
Time        int64
Is_CH       int64
who_CH      int64
Dist_To_CH  float64
ADV_S       int64
ADV_R       int64
JOIN_S      int64
JOIN_R      int64
SCH_S       int64
SCH_R       int64
Rank        int64
DATA_S      int64
DATA_R      int64
Data_Sent_To_BS int64
dist_CH_To_BS float64
send_code   int64
Expanded Energy float64
Attack type object
dtype: object
```

► Summary Info:

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 374661 entries, 0 to 374660
Data columns (total 19 columns):
#   Column              Non-Null Count  Dtype
---  -
0   id                  374661 non-null  int64
1   Time                374661 non-null  int64
2   Is_CH               374661 non-null  int64
3   who_CH              374661 non-null  int64
4   Dist_To_CH          374661 non-null  float64
5   ADV_S               374661 non-null  int64
6   ADV_R               374661 non-null  int64
7   JOIN_S              374661 non-null  int64
8   JOIN_R              374661 non-null  int64
9   SCH_S               374661 non-null  int64
10  SCH_R               374661 non-null  int64
11  Rank                374661 non-null  int64
12  DATA_S             374661 non-null  int64
13  DATA_R             374661 non-null  int64
14  Data_Sent_To_BS     374661 non-null  int64
15  dist_CH_To_BS       374661 non-null  float64
16  send_code           374661 non-null  int64
17  Expanded Energy     374661 non-null  float64
18  Attack type         374661 non-null  object
dtypes: float64(3), int64(15), object(1)
memory usage: 54.3+ MB
```



- ▶ Dataset shape Before Dropping: 374661
- ▶ Dataset shape After Dropping: 365788
- ▶ Total Duplicates Dropped: 8873

- ▶ Checking class distribution

- ▶ Counts:

Attack type
Normal 332040
Grayhole 13909
Blackhole 10049
TDMA 6633
Flooding 3157
Name: count, dtype: int64

- ▶ Percentages (%):

Attack type
Normal 90.77
Grayhole 3.80
Blackhole 2.75
TDMA 1.81
Flooding 0.86

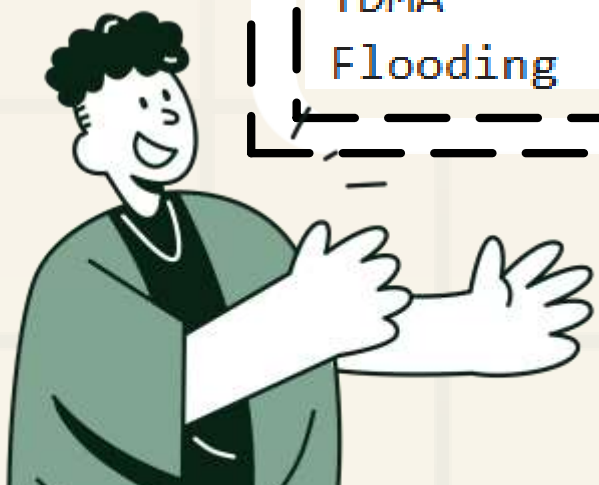
Number of Missing values -
0

Unique Classes before
encoded: ['Blackhole'
'Flooding' 'Grayhole'
'Normal' 'TDMA']

Unique Classes After
encoded: **[0 1 2 3 4]**

DATA PRE-PROCESSING

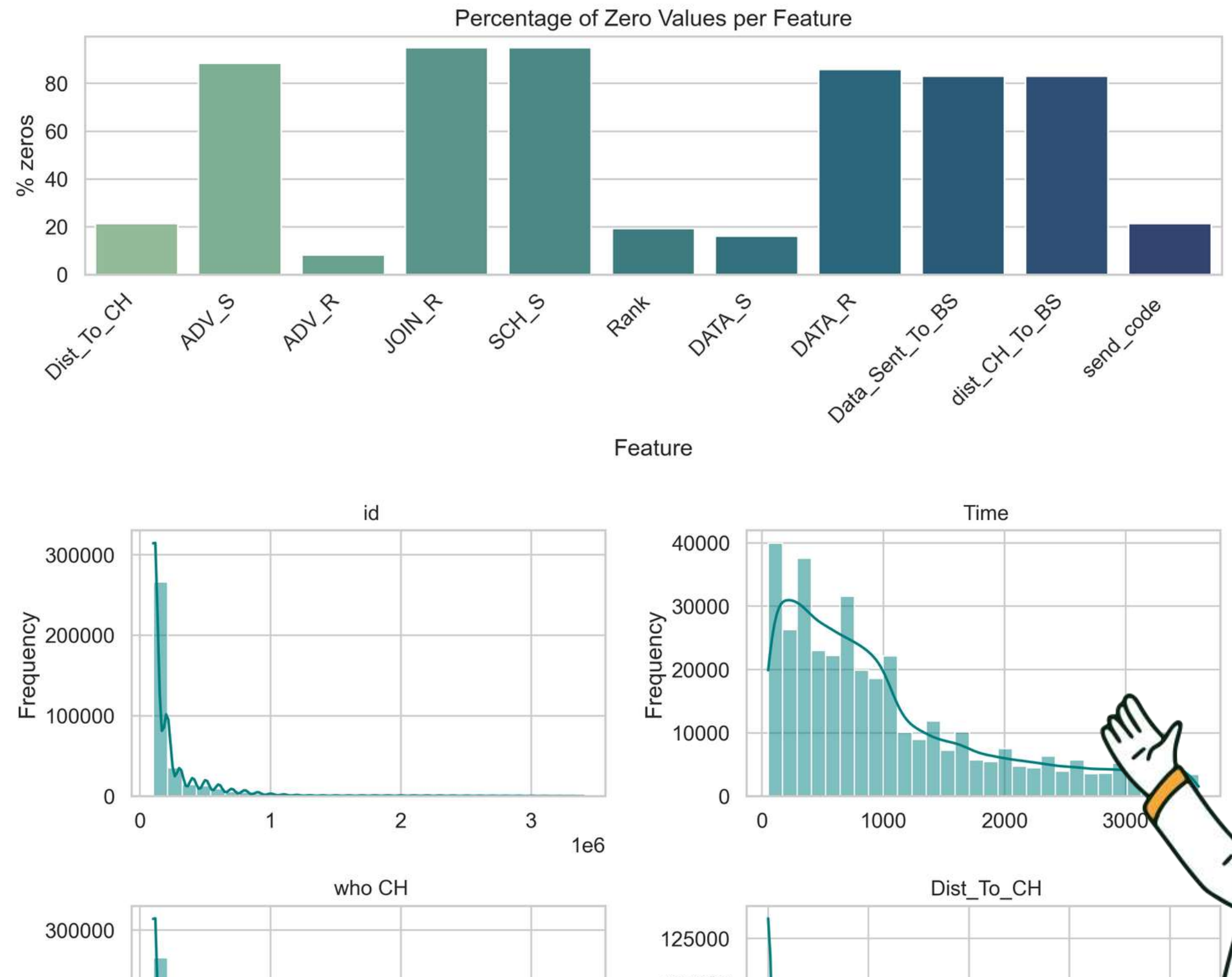
- Checking Missing Values
- Handling Duplicate Values
- Unique values of the target column
- Class Distribution of the target
- Correlation between features
- Profile report

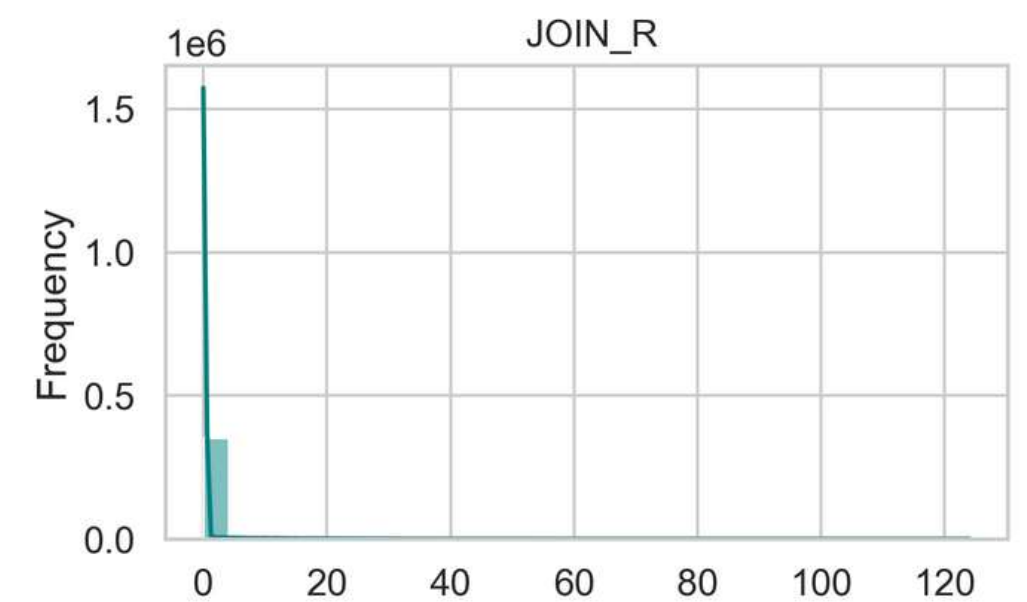
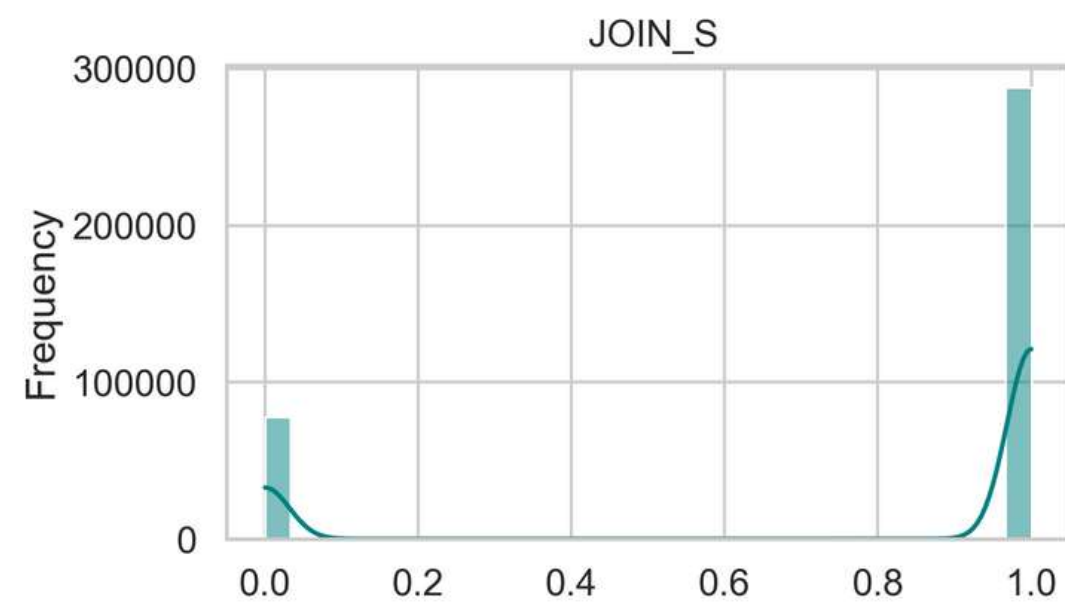
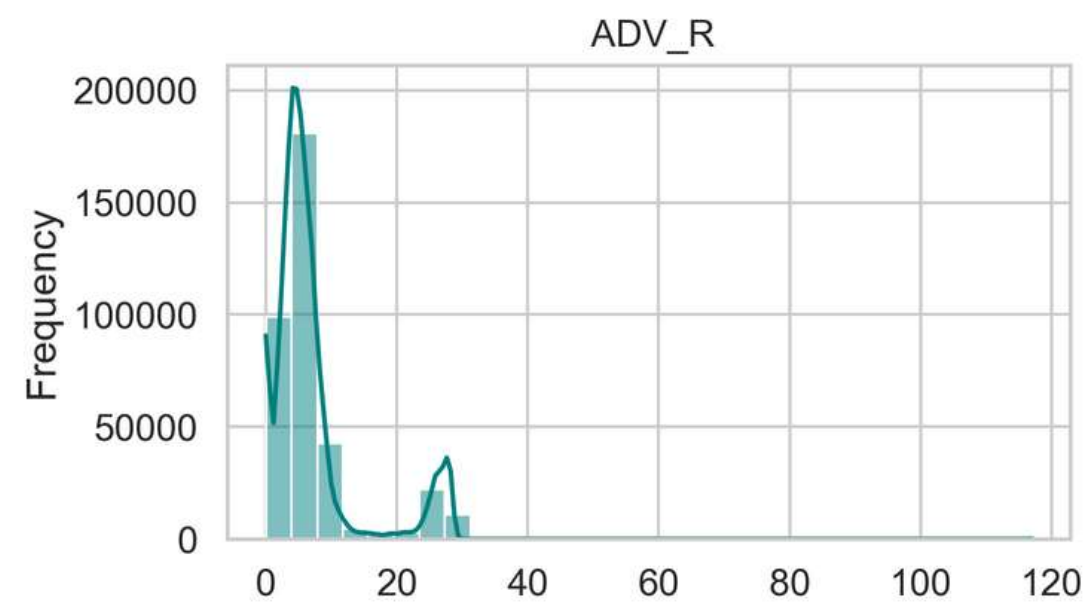
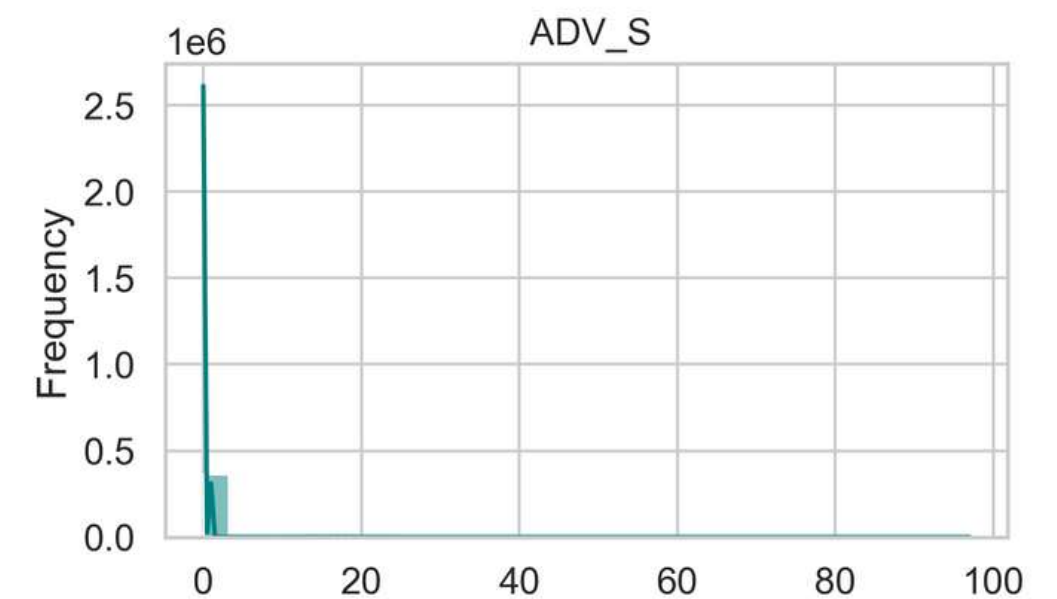
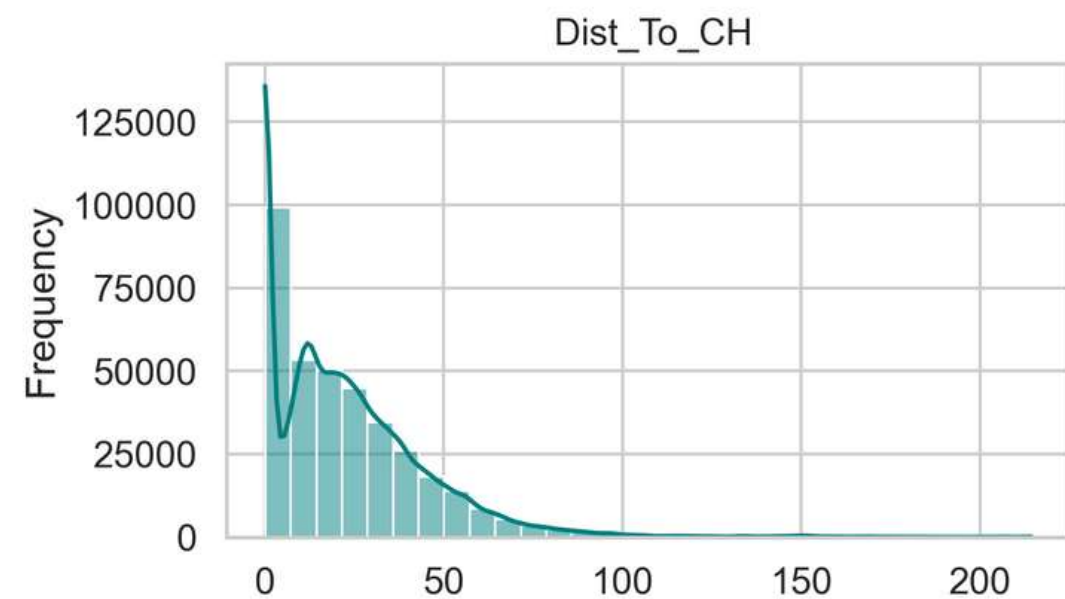
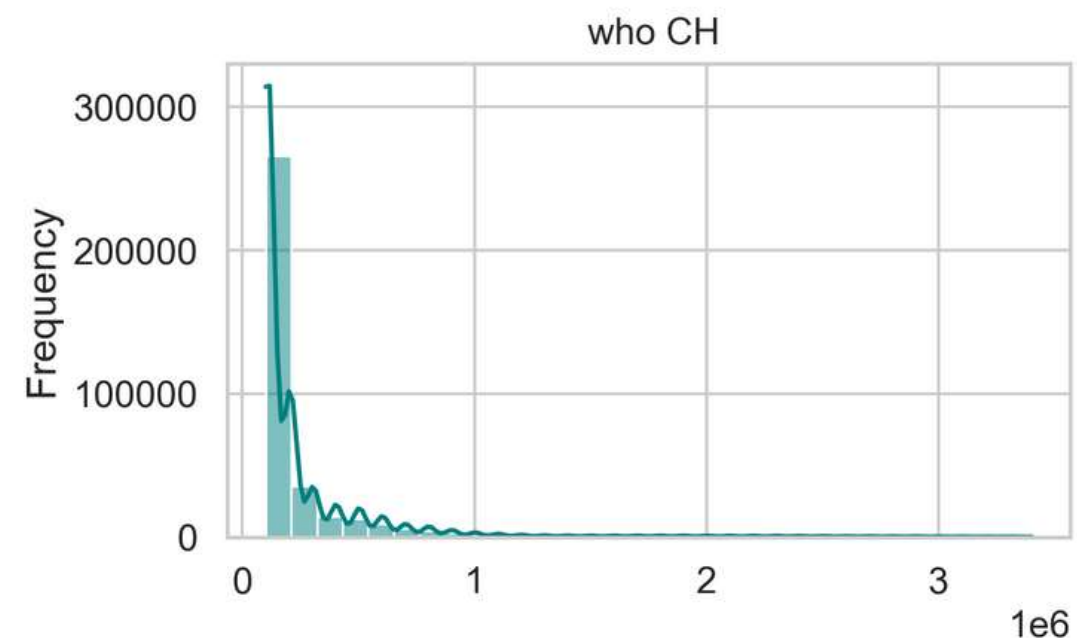
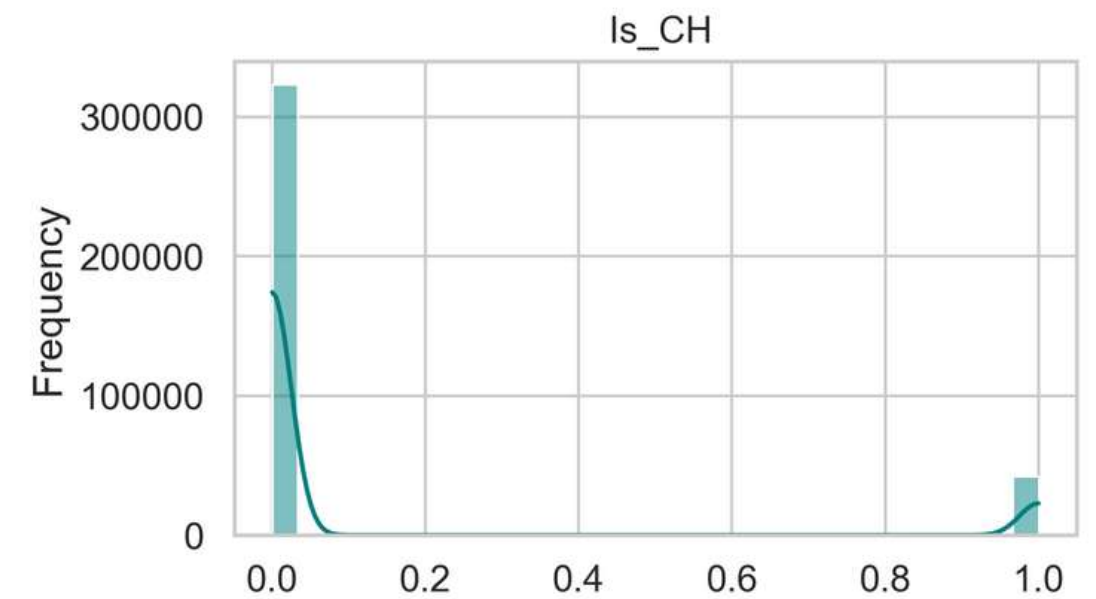
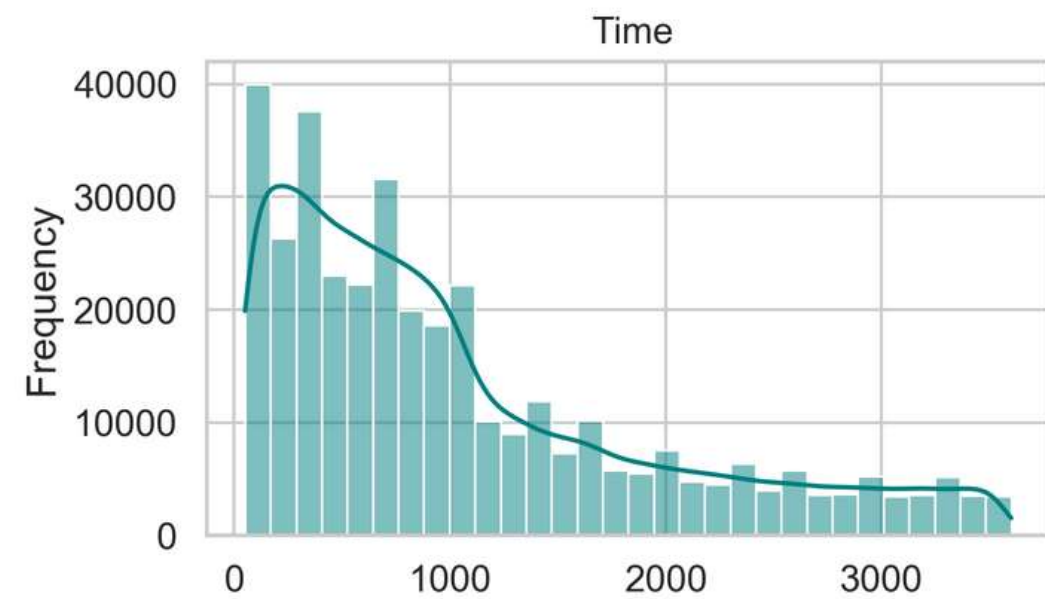
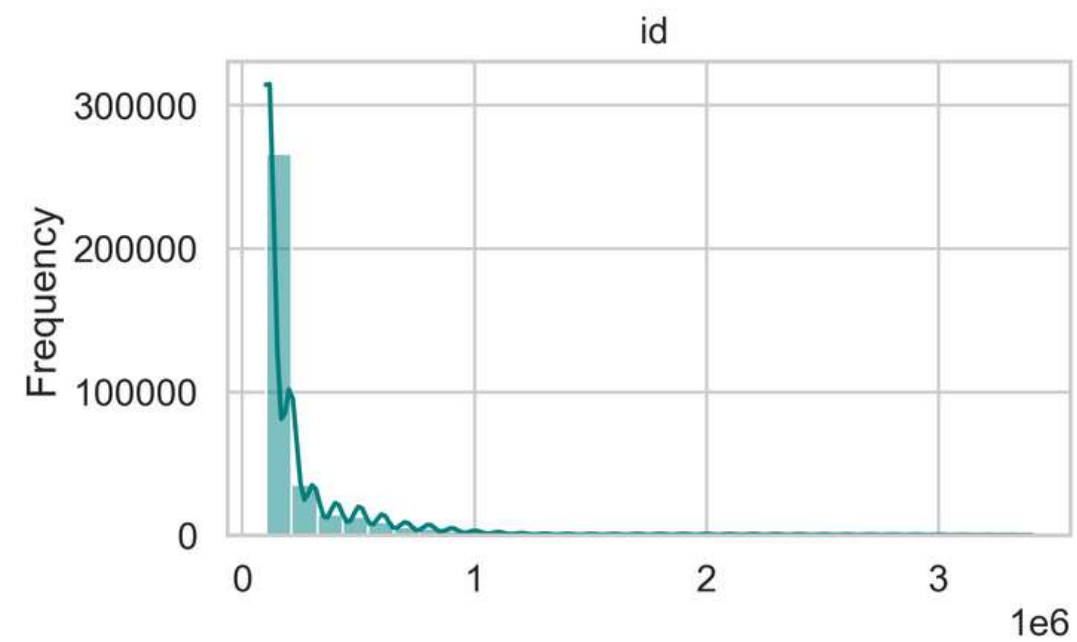


EXPLORATORY DATA ANALYSIS (EDA)

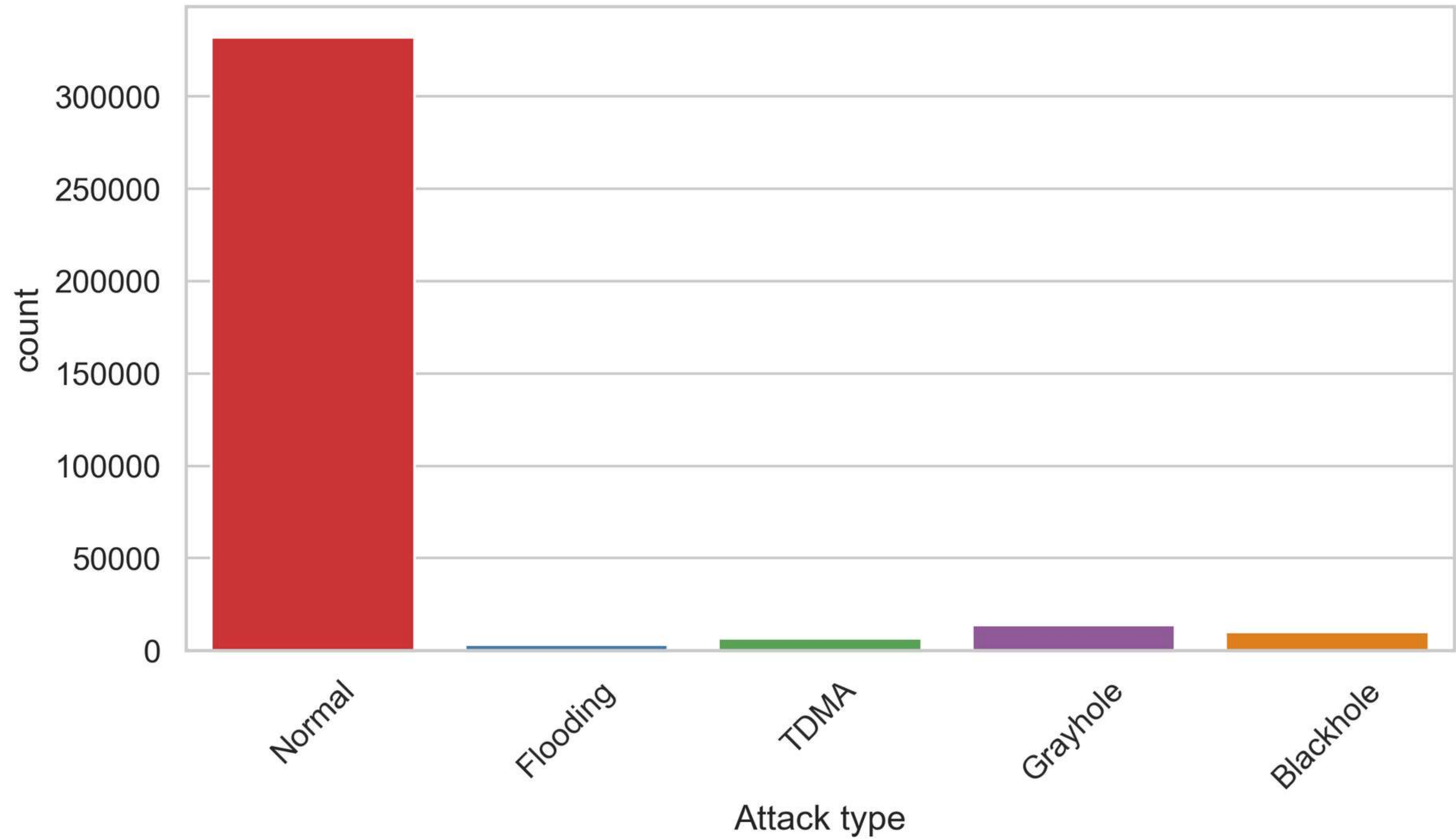
- Univariate
- Bivariate
- Multivariate

First, we see univariate

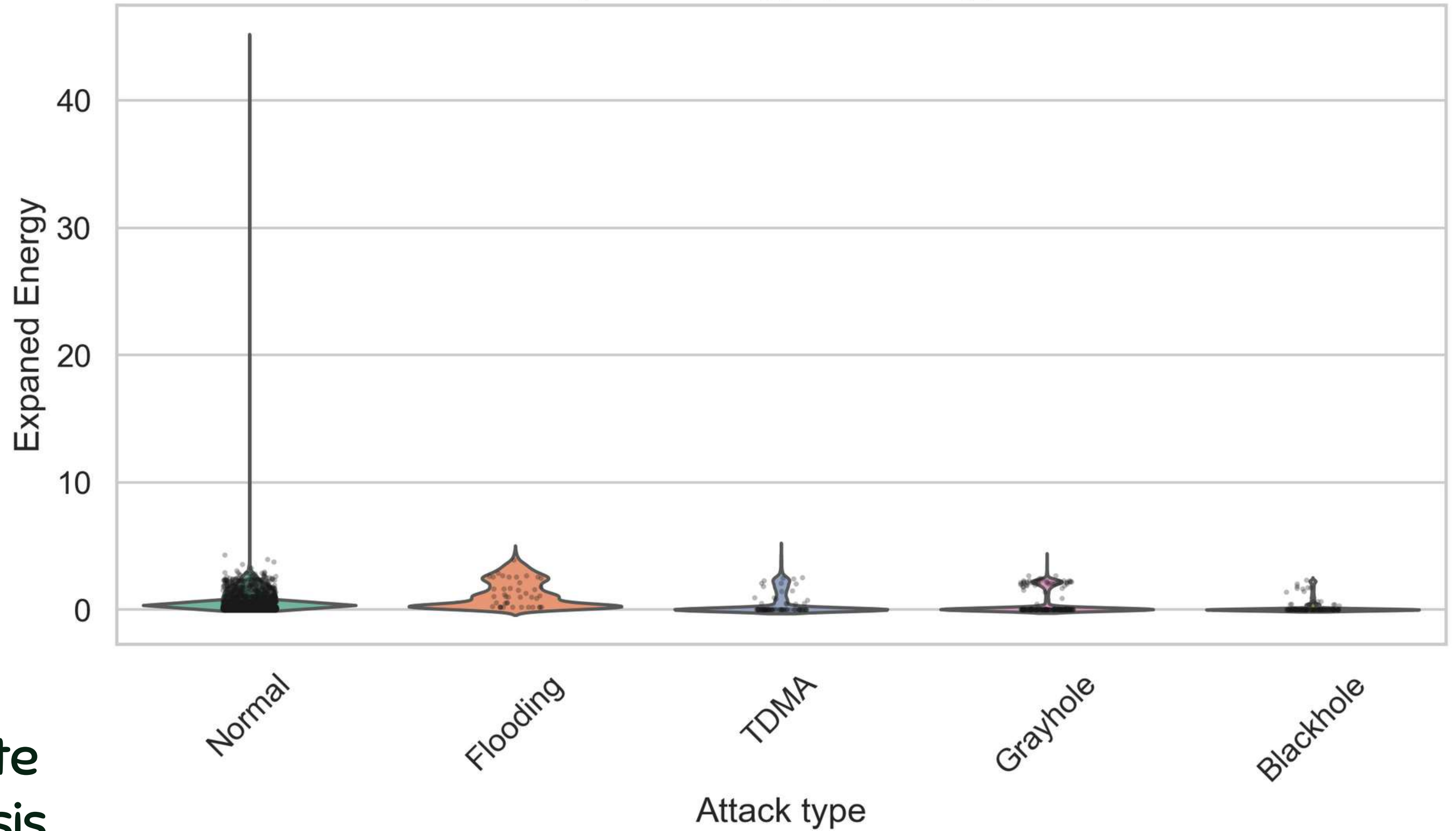




Class Balance

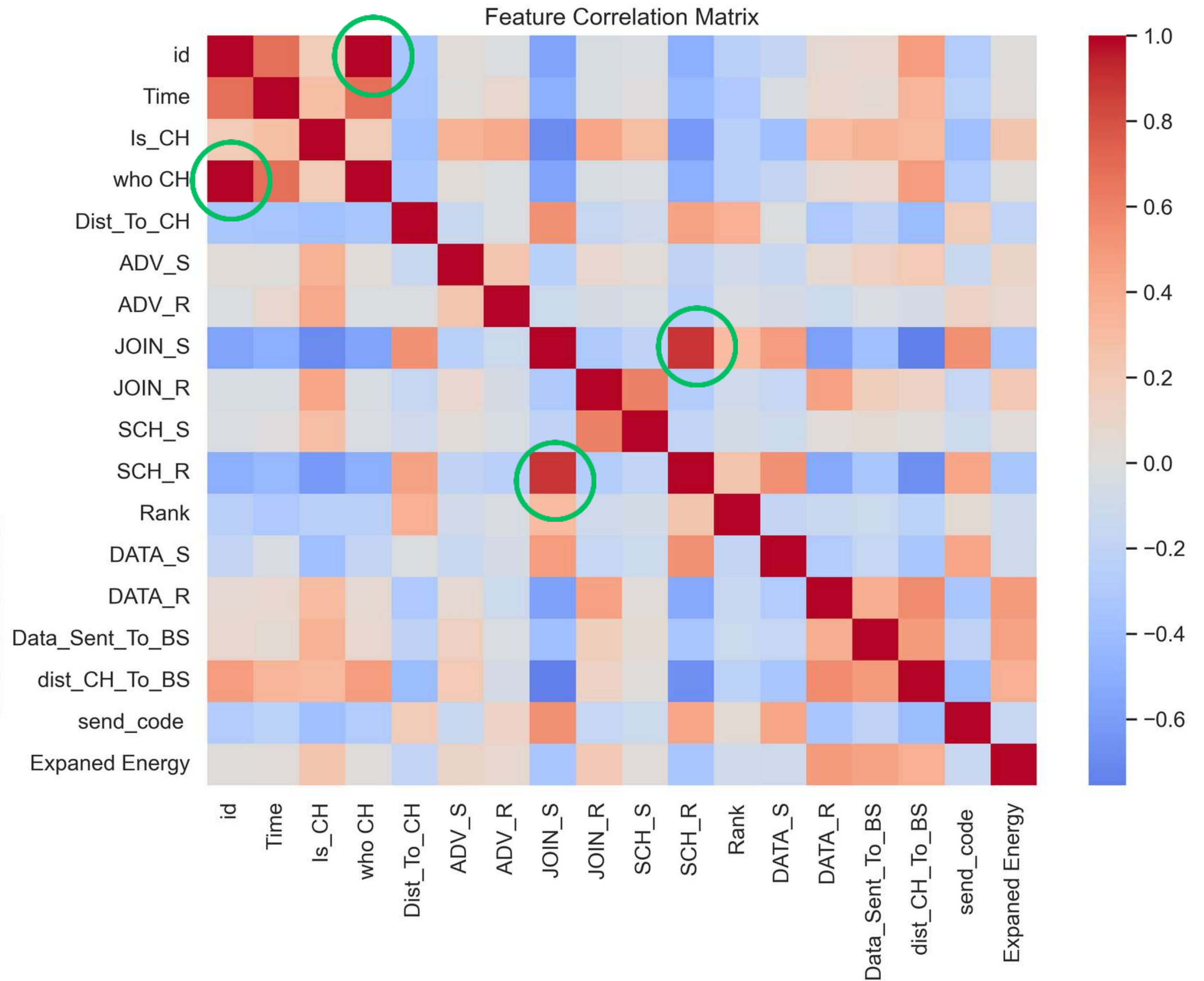


Expanded Energy by Attack type



Bivariate
Analysis

- who CH
- id
- JOIN_S
- SCH_S



```
X_train shape: (292630, 16)
X_test shape: (73158, 16)
Y_train shape: (292630,)
Y_test shape: (73158,)
```

Dropped id, who CH, target column
in X_train and X_test

Unique Classes before encoded:
['Blackhole' 'Flooding' 'Grayhole'
'Normal' 'TDMA']

Unique Classes After encoded: [0 1 2
3 4]

FEATURE ENGINEERING & SCALING

- Correlation-based feature drop
- – $|\text{corr}| > 0.9$
- Implemented MinMaxScalar
- Used LabelEncoder for encoding y features
- Did Train/Test Split 80:20





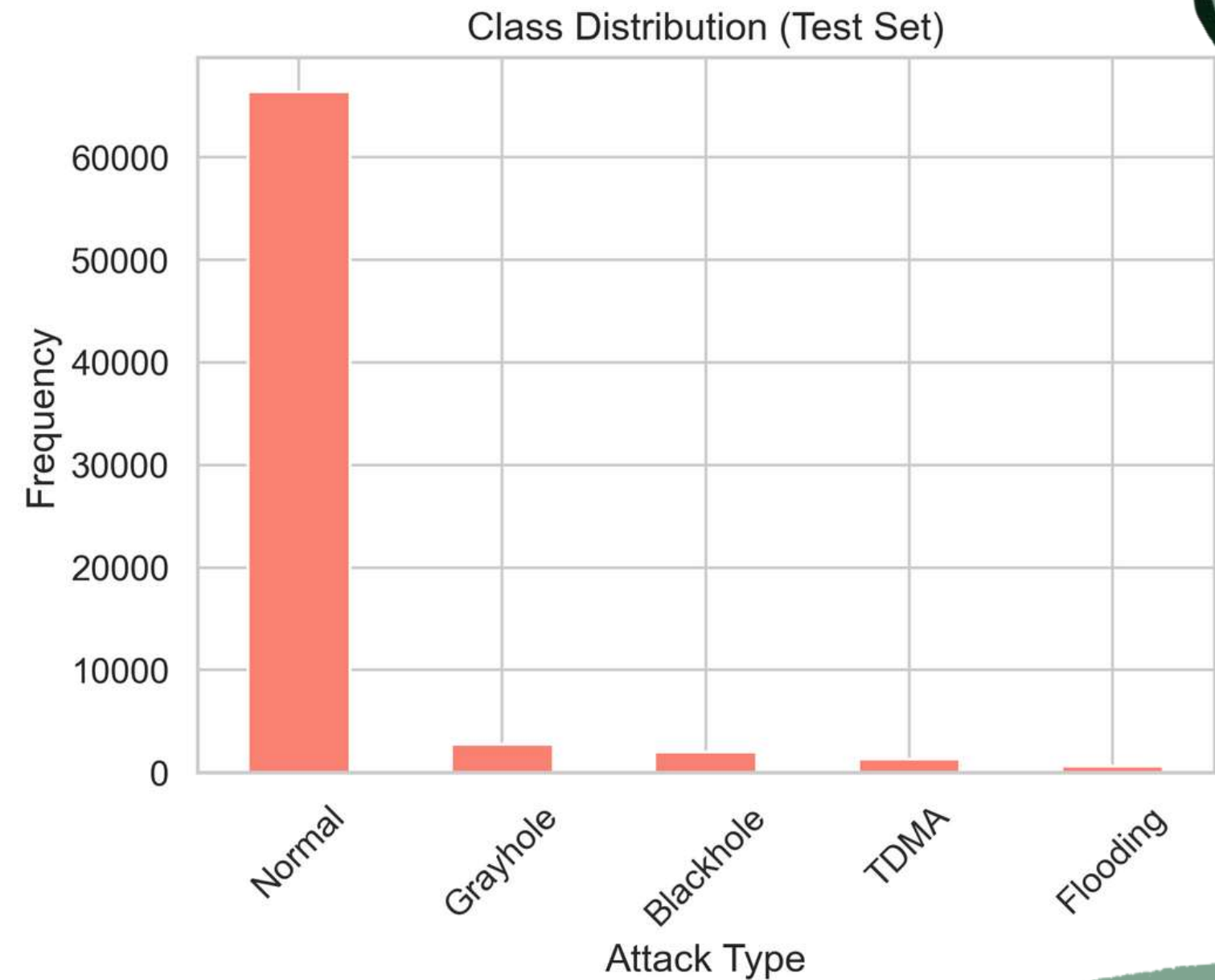
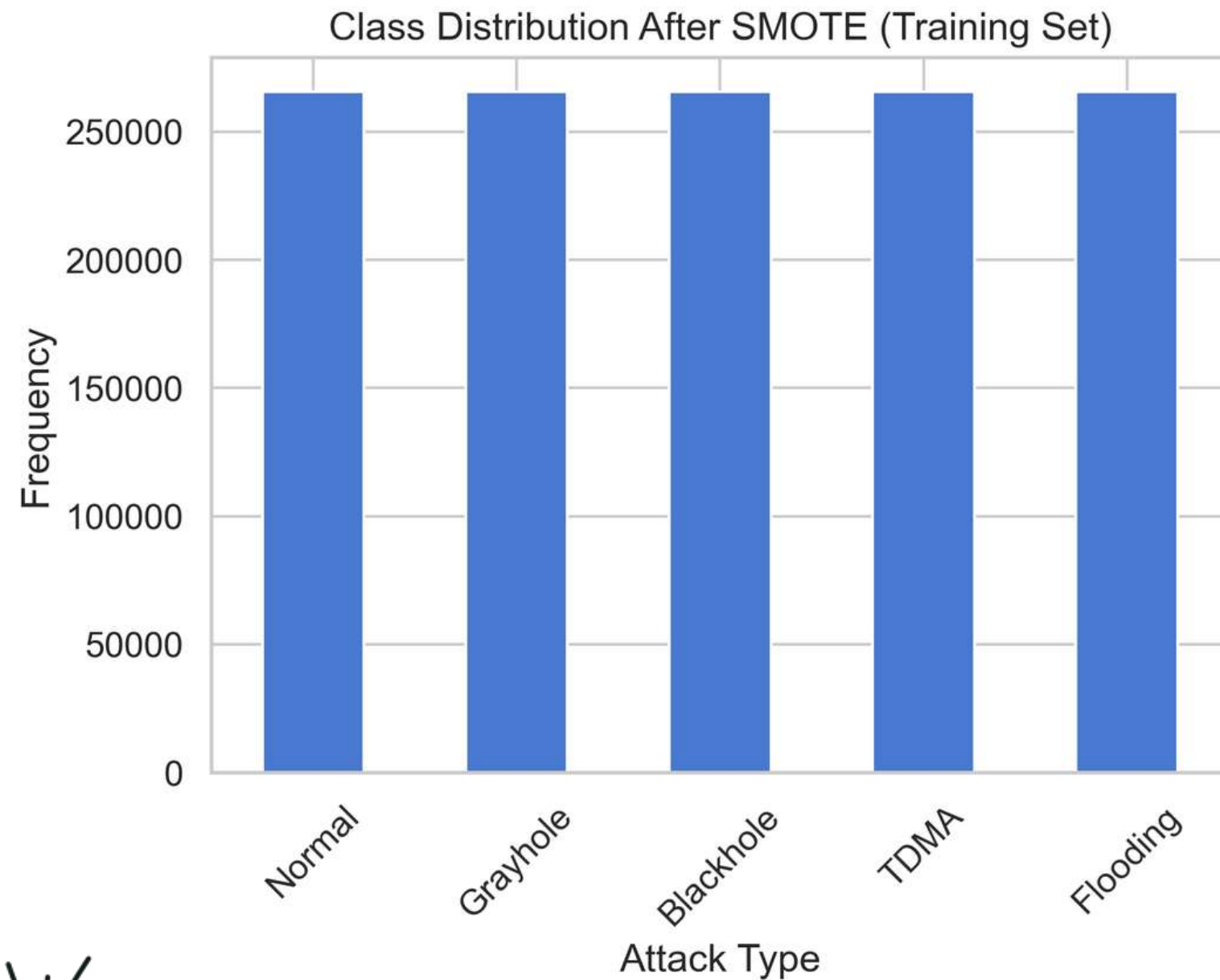
FEATURE SELECTION



- VarianceThreshold: removed low-variance features (< 0.01)
- SelectKBest (ANOVA F-test): top 10
- RFE w/ LogisticRegression: final 10 features
- Selected features: ['Time', 'Is_CH', 'Dist_To_CH', 'JOIN_S', 'SCH_R', 'Rank', 'DATA_S', 'DATA_R', 'dist_CH_To_BS', 'send_code']

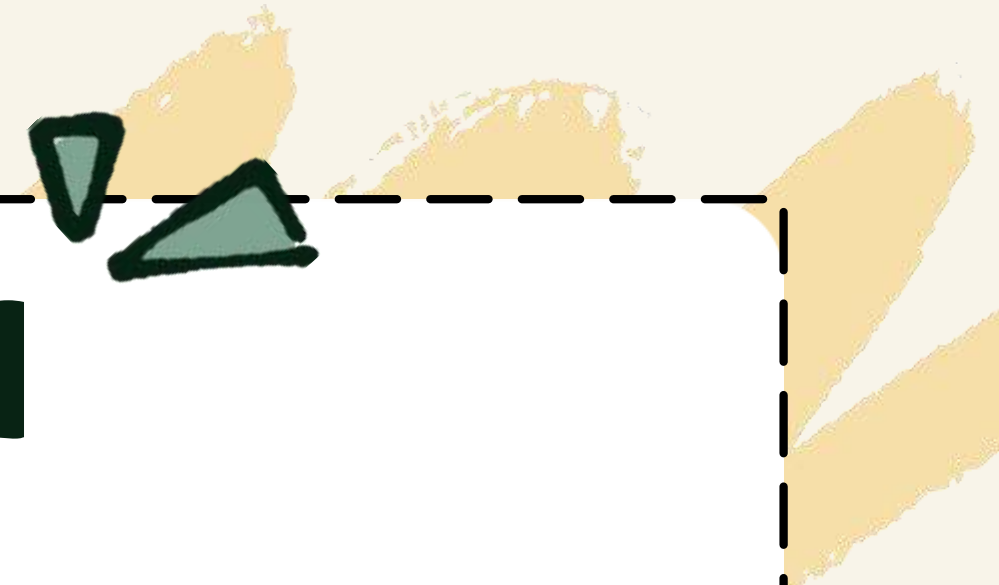


HANDLING CLASS IMBALANCE (SMOTE)



Imbalance handling: SMOTE, SMOTE-Tomek, SMOTE-ENN

- Before SMOTE: {'Normal': 90.77%, 'Grayhole': 3.80%, 'Blackhole': 2.747%, 'TDMA': 1.813%, 'Flooding': 0.863%}
- After SMOTE: {'Normal': 20%, 'Blackhole': 20%, 'Grayhole': 20%, 'TDMA': 20%, 'Flooding': 20%}



MODEL IMPLEMENTATION

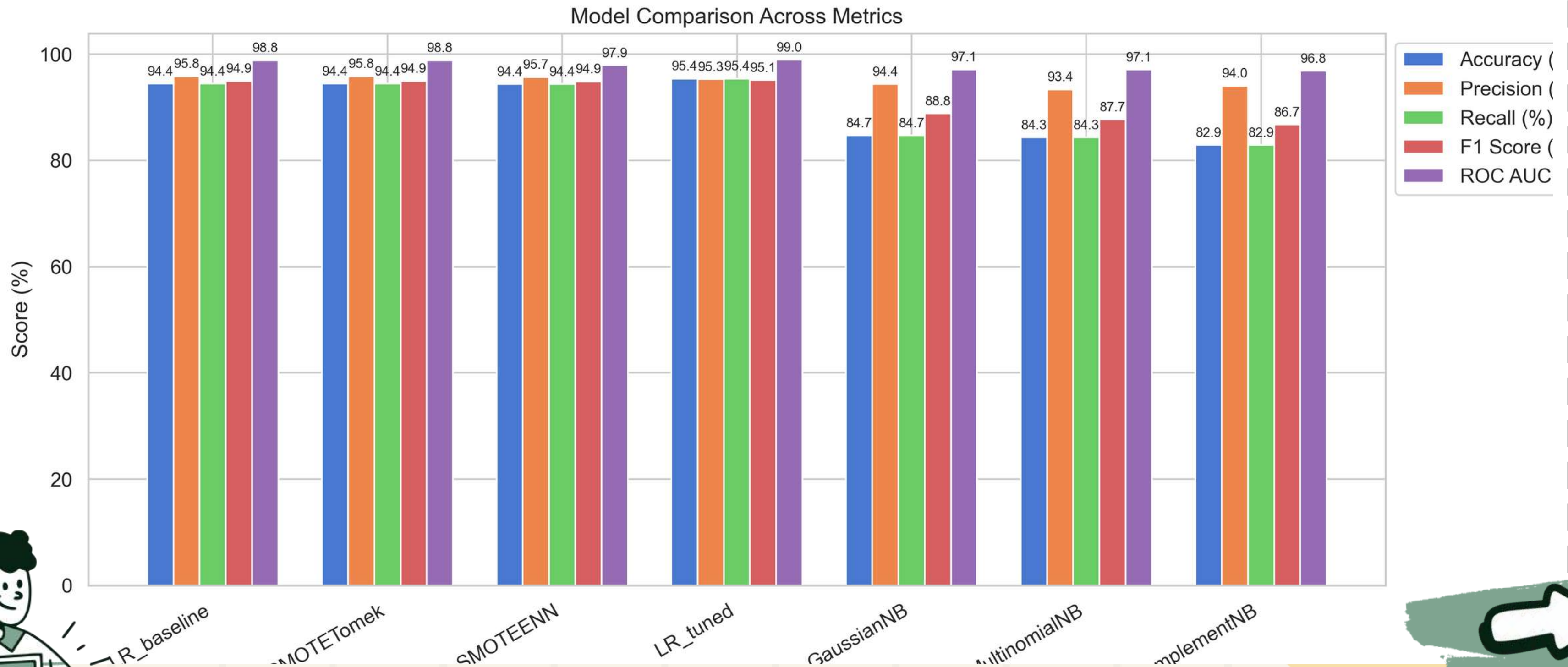
- Logistic Regression variants
 - Baseline
 - SMOTE-Tomek
 - SMOTE-ENN
 - GridSearch-tuned
- Naive Bayes variants
 - GaussianNB
 - MultinomialNB
 - ComplementNB

Hyper-parameter tuning: GridSearchCV on LR ($C \in \{0.01, 0.1, 1, 10\}$)

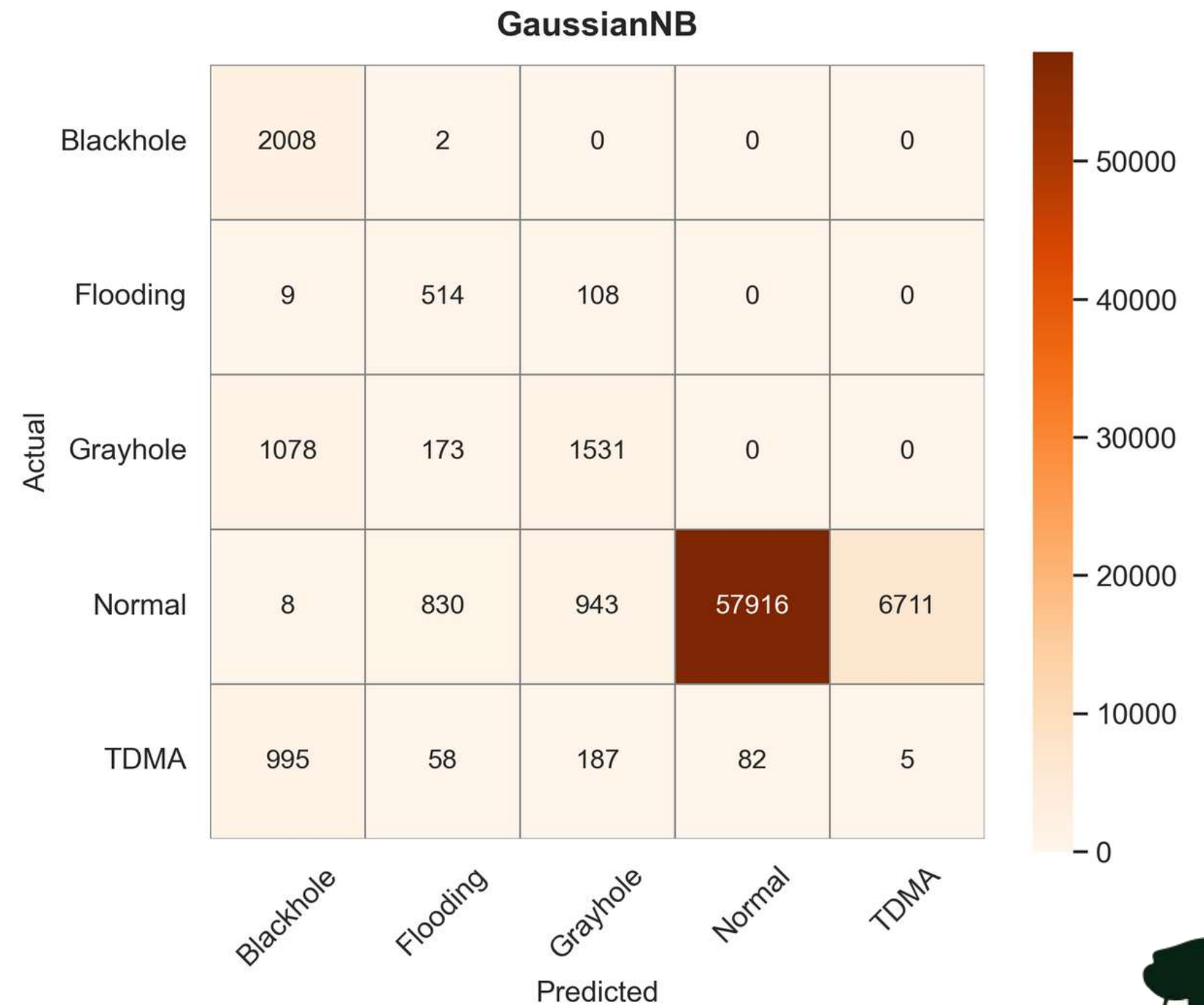
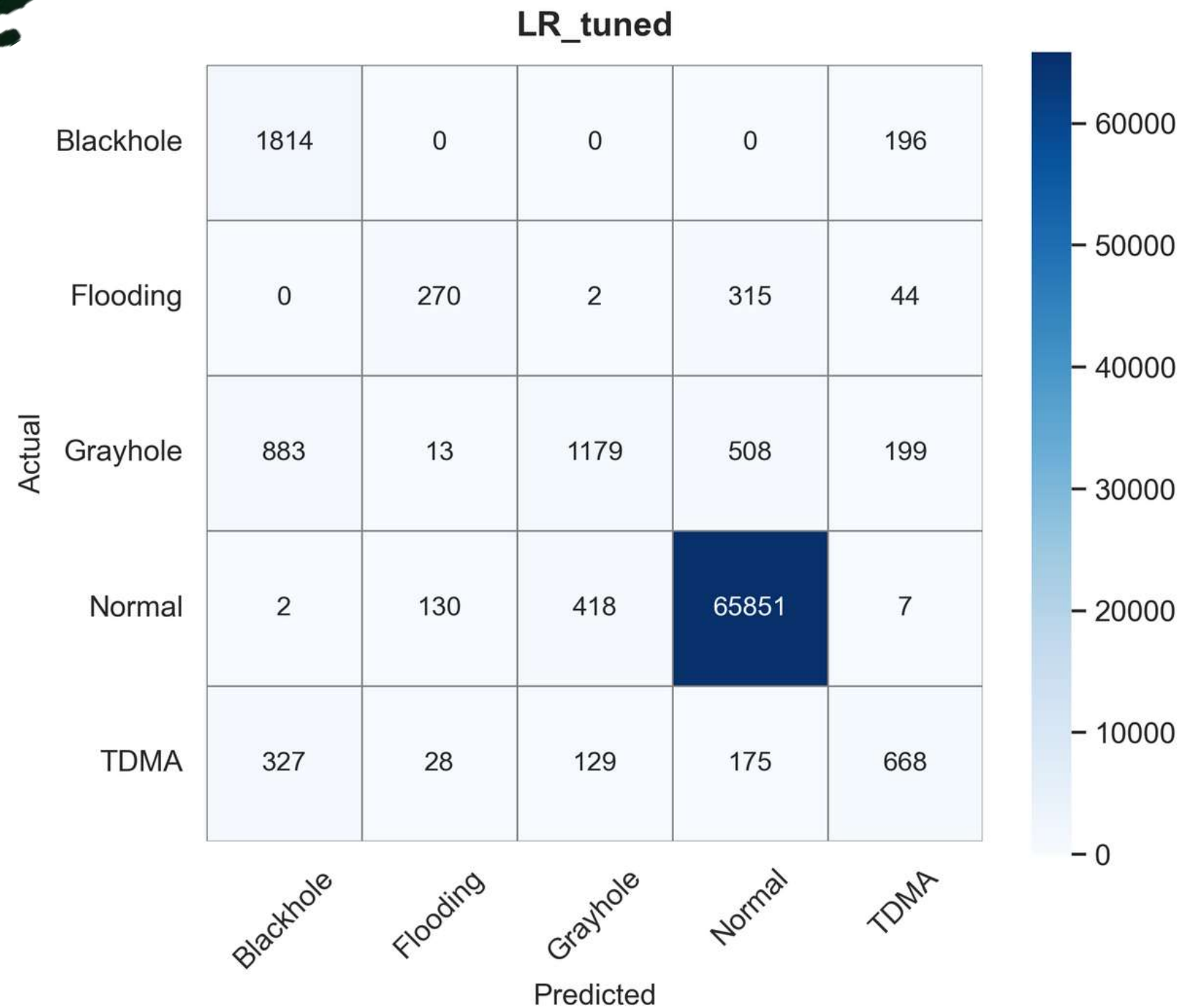


MODEL COMPARISON & SELECTION

Metrics: Accuracy, Precision, Recall, F1-score, ROC-AUC



TWO BEST MODELS

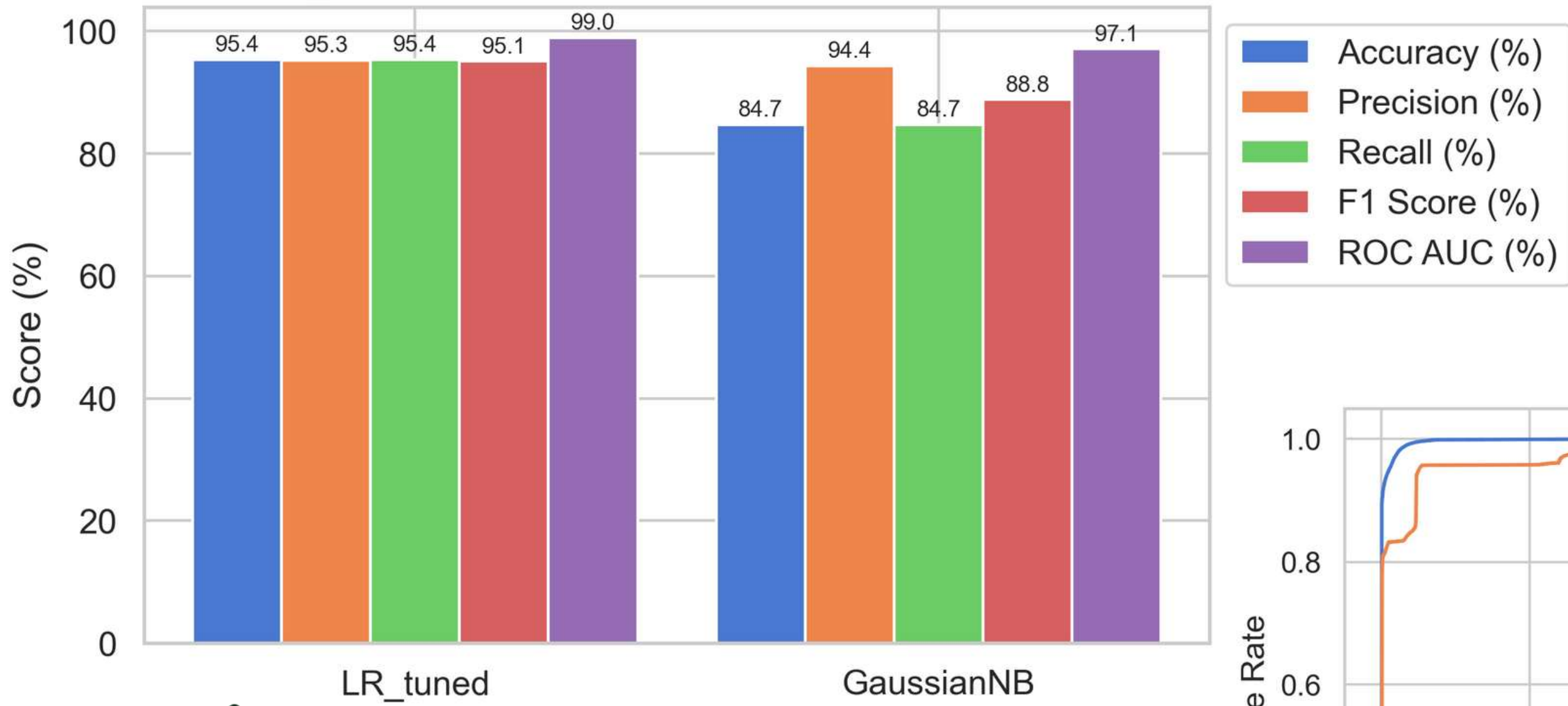


LR_tuned: highest overall F1 & AUC

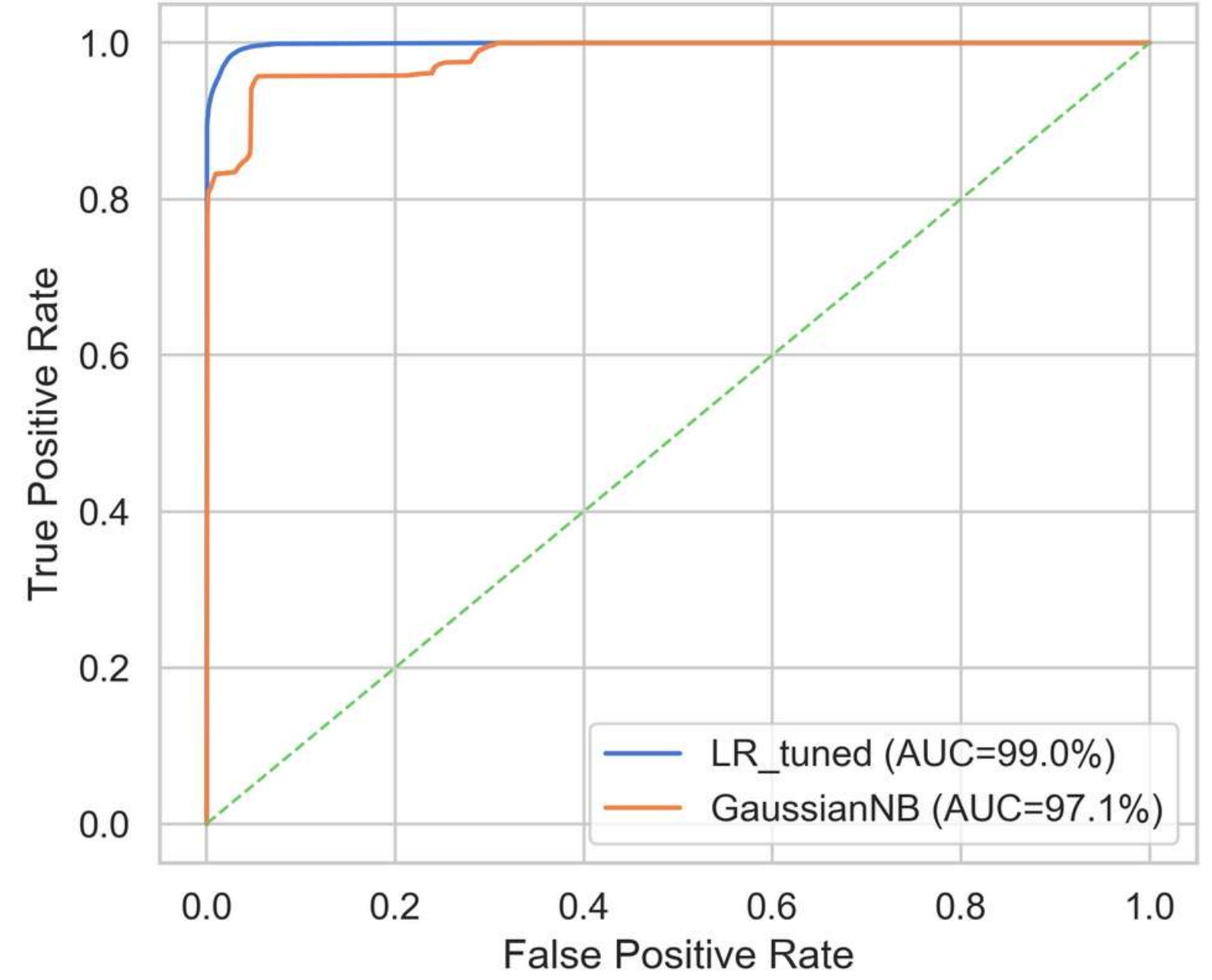
GaussianNB: lightweight, strong probabilistic performance



LR_tuned vs GaussianNB — Performance Metrics



ROC Curves Comparison



COMPARSION & ROC_CURVE





**THANK YOU FOR
LISTENING!**

